VIGNESH INFRA DEVELOPERS PRIVATE LIMITED

Company code: VID

LogicDWG Tools

Setup and usage guide for each tool

Company	**VIGNESH INFRA DEVELOPERS PRIVATE LIMITED**
Company code	**VID**
Platform	**AutoCAD / Civil 3D 2026**
Package	**vidLoader 1.1.4 | LogicDWG dialog 1.1.9**
Tools covered	**14 commands**

Contents

1. Installation and loading (all tools)

2. LogicDWG dialog and Map Zone

3. Survey Points (VIDDGPSTOSP)

4. Rail Track export (VIDLINETOEXPORTLEVEL)

5. Rail Track generate (VIDDGPSTOLINE)

6. Chainage (VIDCHAINAGERUNNER)

7. OHE (VIDDGPSTOOHE)

8. Building (VIDDGPSTOBOX)

9. Gradient generate (VIDGRADIENT)

10. Gradient align (VIDGRADIENTALIGN)

11. Gradient clean up (VIDGRADIENTCLEANUP)

12. Delete CAD layers (VIDCADMERGELAYERS)

13. Dim settings (VIDDIMSETTINGS)

14. Convert dim unit (VIDDIMMTOFT)

15. CSV formats and troubleshooting

1. Installation and loading (all tools)

All tools are AutoLISP files loaded by one loader. You set this up once, then every tool is available from the LogicDWG dialog or by typing its command.

What is in the package

File / folder	Purpose
vidLoader.lsp	Loads every tool file and prints a status list ([OK] / [ERROR] / [MISSING]).
tools\logicDwg.lsp + logicDwg.dcl	The main button dialog (command VIDLOGICDWG).
tools*.lsp	One file per tool.
tools\vid_blocks_tools.dwg	Block library. Needed for Chainage symbol and Gradient blocks.
tools\vidGradient_template.csv	Example CSV for the Gradient tool.

Setup

1. Unzip the package. Keep **vidLoader.lsp** and the **tools** folder together (the tools folder sits next to the loader).
2. Recommended: copy both into your Civil 3D support folder, so the result is `...\Support\vid\vidLoader.lsp` and `...\Support\vid\tools\`. The default path searched is `%APPDATA%\Autodesk\C3D 2026\enu\Support\vid\tools`.
3. Add that folder to **Options > Files > Support File Search Path**, and to **Trusted Paths** (TRUSTEDPATHS) so AutoCAD does not block the files.
4. Auto-load (optional): in **APPLOAD > Startup Suite > Contents**, add **vidLoader.lsp**.
5. Without auto-load: type **APPLOAD**, pick vidLoader.lsp and click Load (or drag and drop it into the drawing).

How to use

1. After loading, read the command line. It lists each file as [OK] and ends with *Loaded : 14 / 14* and *LogicDWG is ready*.
2. Type **VIDLOGICDWG** (or LOGICDWG) to open the button dialog. Or type any tool command directly (listed on each page).
3. If a file shows [ERROR] or [MISSING], the loader retries it form by form and prints the failing line. Check the tools folder is complete.

Block DWG lookup: Chainage and Gradient find `vid_blocks_tools.dwg` in the same folder as their LSP, then in the Support\vid\tools folders, Trusted Paths and the drawing folder. If it is not found, the command lists the places it checked.

2. LogicDWG dialog and Map Zone

`VIDLOGICDWG / LOGICDWG / VIDMAPZONE`

The dialog is a control panel. Each button runs one tool. Map Zone buttons set the coordinate system and switch on the online map.

Dialog layout

Section	Button	Runs
Map Zone	43 / 44 / Off	Sets UTM84-43N or UTM84-44N, turns GeoMap Hybrid on and zooms extents. Off turns GeoMap off.
Generate from CSV	Survey points: Generate	VIDDGPSTOSP
	Rail track: Export / Generate	VIDLINETOEXPORTLEVEL / VIDDGPSTOLINE
	Chainage: Generate	VIDCHAINAGERUNNER
	Ohe: Generate	VIDDGPSTOOHE
	Building: Generate	VIDDGPSTOBOX
	Gradient: Clean up / Align / Generate	VIDGRADIENTCLEANUP / VIDGRADIENTALIGN / VIDGRADIENT
Settings	Delete Cad Layers: Delete	VIDCADMERGELAYERS
	Dim Settings: Apply	VIDDIMSETTINGS (dialog stays open afterwards)
	Convert Dal unit: Convert	VIDDIMMTOFT

Setup

Load the tools (section 1). No other setup. logicDwg.dcl must stay in the same folder as logicDwg.lsp.

How to use

1. Type **VIDLOGICDWG**. The dialog opens.
2. Set the map zone first if you need the online map: click **43** or **44**. In Civil 3D / Map 3D the coordinate system is assigned automatically. In plain AutoCAD the Geographic Location dialog opens, so choose *WGS84 UTM Zone 43N* (or 44N) and click OK.
3. Click the button for the tool you want. The dialog closes and the tool starts. Only Dim Settings reopens the dialog.

VIDMAPZONE is a command-line version: it asks *Enter UTM Zone [Z43/Z44/Off] <Z43>*.

3. Survey Points

`VIDDGPSTOSP`

Imports a DGPS CSV and draws every survey point as a POINT plus three MTEXT labels (point name, code, elevation).

Setup

- Needs a CSV with columns: **Point Name, Code, Northing, Easting, Elevation**. Optional: **Local Time**.
- Set the drawing to the right UTM zone first (section 2) so the points land on the map.
- Text height is 0.5 drawing units. Points go on a layer you choose in the dialog.

How to use

1. Click **Survey points > Generate** (or type VIDDGPSTOSP).
2. In the **Survey Point Layer** dialog choose **Custom** (all points on one layer you type) or **CSV Code** (one layer per code, named `sp_<Code>`). Click OK.
3. Select the DGPS CSV file.
4. The command line prints a summary and the first five records. Rows with a blank name or non-numeric Northing / Easting / Elevation are skipped.
5. Result: for each valid row a POINT at (Easting, Northing, Elevation), with MTEXT for name, code and elevation. The end report says *Survey record integrity: OK* when the counts match.

Example CSV (Survey Points)

Point Name	Code	Northing	Easting	Elevation	Local Time
1	BM	1234567.120	765432.480	412.350	10:15:02
2	KERB	1234570.305	765436.910	412.402	10:15:40
3	KERB	1234574.880	765441.030	412.388	10:16:12
4	LC	1234581.250	765447.770	412.610	10:17:05

Notes: Layer names are cleaned (spaces and special characters become _). New layers are created in colour 4.

4. Rail Track Export (levels to CSV)

`VIDLINETOEXPORTLEVEL  (old name EXPORTLEVEL20)`

Reads levels along a 3D polyline and writes a Chainage,Elevation CSV at a fixed interval. This CSV is the level source for the Gradient tool.

Setup

- The track must be a **3D polyline** (or polyline with elevation).
- Save the drawing first. The CSV is written to the drawing folder. If the drawing is unsaved or the folder is not writable, a Save dialog appears.

How to use

1. Click **Rail track > Export** (or type VIDLINETOEXPORTLEVEL).
2. Select the polyline.
3. In the **Export Level** dialog enter **Start Chainage** as km/metre in one box, e.g. `21/000` (metre 0 to 999), the **Interval** (whole metres, default 20) and **Track Type** (Up or Down). Click Ok.
4. A file is written automatically, named `<drawing>_<Up|Down>_railLevel_v01.csv`. Running again makes v02, v03 and so on, counted separately for each drawing and track type.
5. A message confirms the file path.

Example output CSV (written by the tool)

Chainage	Elevation
21000.000	1730.339
21020.000	1730.201
21040.000	1730.060
21060.000	1729.918

Notes: Chainage = (km x 1000 + m) + distance along the polyline. A level is written every Interval metres from the start up to the end of the line.

5. Rail Track Generate

`VIDDGPSTOLINE`

Turns DGPS points of each Code into polylines or 3D polylines. It also detects hectometre posts (HP) automatically and uses them to give every track the same direction as the chainage.

Setup

- CSV columns required: **Point Name, Code, Northing, Easting, Elevation**. Optional: **Local Time**.
- HP rows are found by name or code, e.g. `HP 71/600, HP 71-600, HP 71_600, HP 71 600, HP71/600`. Need at least 2 HP points for the direction step.
- Set the UTM zone first (section 2).

How to use

1. Click **Rail track > Generate** (or type VIDDGPSTOLINE) and select the CSV.
2. **Rail Track - Select Layer** dialog: choose **Polyline** or **3D Polyline**, then pick the Code(s) from the list. Codes R, R1, R2 ... R<n> are pre-selected.
3. Set **Options**: **Angle filter +/-** (whole number of degrees, default 5) removes spike vertices. Tick **Use Local Time column** to join points in time order; leave it off to join nearest points from both ends of the chain.
4. Click **Select Layers**. The **Layer Options** dialog opens: **Default layers** (pick from the list, e.g. 1-TRACK, 1-OHE, 1-PLATFORM), **From CSV** (layer `1-TRACK_<Code>` per code) or **Custom** (type a name). Click OK.
5. The report shows rows, valid, unique codes and the points per code. One polyline is created per code. The temporary HP curve and HP layer are removed afterwards.

Example CSV (Rail Track Generate)

Point Name	Code	Northing	Easting	Elevation	Local Time
101	R1	1234500.110	765400.250	412.100	09:30:01
102	R1	1234503.440	765404.870	412.095	09:30:03
103	R1	1234506.790	765409.420	412.090	09:30:05
201	R2	1234498.020	765402.980	412.110	09:41:12
HP 71/600	HP	1234510.000	765415.000	412.080	09:50:00

Notes: Groups with fewer than 2 points are skipped. Last-used options are remembered between runs. The Chainage and OHE tools expect tracks on **1-TRACK** / **1-track** (OHE) so use the Default layer 1-TRACK for rails.

6. Chainage

`VIDCHAINAGERUNNER`

Places chainage marks (dot + text) along selected polylines at a set increment. Optionally adds a KM arrow symbol block at the end of each polyline.

Setup

- Optional symbol needs `vid_blocks_tools.dwg` with the block `vid_kmArrow_001` (the default name). If the DWG is not found the run continues without symbols.
- Everything is drawn on layer **1-CHAINAGE** (created automatically).

How to use

1. Click **Chainage > Generate** (or type VIDCHAINAGERUNNER).
2. Select one or more polylines. Each runs separately with the same start value.
3. In the **Chainage Runner** dialog choose **Forward** (value is at the polyline start) or **Backward** (value is at the end; chainage decreases).
4. Enter **KM / metre** as e.g. `21/000`, and the **Increment** in metres (default 100).
5. Tick **Add Symbol** if wanted and set **Symbol Name**, **Symbol Height** (distance of the symbol from the polyline, default 120) and **Direction** (Up = left of the polyline, Down = right). Click Ok.
6. Result: a filled circle (0.3 radius) and MTEXT label (height 2, e.g. `21/100`) at each step, with km roll-over. The symbol shows *KM 22* on top and *(21/999.85M)* below.

Notes: Select the polyline in the direction of increasing chainage. The final message gives the polyline, mark and symbol counts.

7. OHE

`VIDDGPSTOOHE`

Reads OHE-type points from a DGPS CSV and draws a small 0.3 square beside the nearest track plus a stacked-fraction label, for each mast / portal.

Setup

- CSV columns required: **Code, Northing, Easting**. Header can be anywhere near the top.
- Codes must start with **OHE**, **OH**, **OM** or **PORTAL**. The rest of the code is the identifier, e.g. `OH 40/22A`. Separators - . and a space are converted to /.
- Track polylines must exist on layer **1-track** (draw them with Rail Track Generate first).

How to use

1. Click **Ohe > Generate** (or type VIDDGPSTOOHE) and select the CSV.
2. At the prompt *Track polylines [Auto/Select] <Auto>* press Enter to use every polyline on 1-track, or type S and select them.
3. The tool finds the nearest track for each point and builds the square on the side **away from the track** (left-hand point = left, right-hand = right).
4. A stacked MTEXT label (e.g. 40 over 22A) is placed on the rectangle edge, rotated to the track direction and kept readable.
5. Everything goes on layer **1-OHE-CSV**. The report lists valid rows, rectangles, texts and failures.

Example CSV (OHE)

Point Name	Code	Northing	Easting	Elevation
31	OHE 40/22A	1234512.340	765420.110	412.700
32	OH 40-24	1234530.870	765441.560	412.690
33	PORTAL 41/02	1234552.410	765463.020	412.720
34	OM 41/04	1234570.150	765484.330	412.680

Notes: Square size 0.3 and text height 1.5 are set at the top of the LSP if you want to change them.

8. Building

`VIDDGPSTOBOX`

Builds one perfect rectangle from the 3 or 4 DGPS points of each Code group (rooms, buildings, sheds, etc.) and labels it with the code.

Setup

- CSV columns required: **Code, Northing, Easting, Elevation**. Optional: **Local Time** (sorts points inside a group).
- Optional: curves on layer **1-TRACK**. The label is rotated to follow the nearest one.

How to use

1. Click **Building > Generate** (or type VIDDGPSTOBOX) and select the CSV.
2. **Dgps To Box - Select Layers**: tick the Code groups to build (room, building, toilet, station, shed, office, quarter, shop and similar names are pre-selected). Click Select Layers.
3. **Layer Options**: choose a default layer (e.g. 1-BUILDINGS), **From CSV** (`1-TRACK_<Code>`) or a custom name. Click OK.
4. A closed rectangle is made per group: point 1 to 2 sets direction and length, point 3 sets the width; the 4th corner is computed. A group with more than 4 points is split into sets of 4 (or 3).
5. An MTEXT label (height 0.5) is placed at the centre on the same layer. The report gives rectangles created, failed and points not used.

Example CSV (Building)

Point Name	Code	Northing	Easting	Elevation	Local Time
51	Motor Room	1234600.100	765500.200	413.000	11:02:10
52	Motor Room	1234600.140	765506.220	413.010	11:02:25
53	Motor Room	1234596.150	765506.250	413.005	11:02:40
54	Motor Room	1234596.110	765500.230	413.000	11:02:55

Notes: Groups with fewer than 3 points are skipped and reported. The CSV word APANDENT is written as ABANDONED BUILDING.

9. Gradient Generate

`VIDGRADIENT`

Places a gradient block at each position along a track and fills its attributes (chainage, rail level, line, tail and head slope) from a CSV.

Setup

- CSV header: `G_POSITION,G_CHAINAGE,G_RAILLEVEL,G_LINE,G_TAIL,G_HEAD,G_NOTE`. A tab-separated file also works. A ready example is `vidGradient_template.csv`.
- **G_POSITION** = distance in metres along the selected polyline. The G_NOTE column may contain commas.
- `vid_blocks_tools.dwg` must contain the gradient block (default name `vid_gradient_001`) with attributes G_POSITION, G_CHAINAGE, G_RAILLEVEL, G_LINE, G_TAIL, G_HEAD, G_NOTE and dynamic properties G_POSITION, G_TAIL_ROT, G_HEAD_ROT.

How to use

1. Click **Gradient > Generate** (or type VIDGRADIENT).
2. *Select Rail Track*: click the polyline. Its length is printed.
3. Select the CSV file.
4. Enter the block name (Enter = vid_gradient_001) and the **Gradient Height** (Enter = 80), the offset of the block to the right of the track.
5. The block is imported from the DWG (replacing an older definition), then one block is inserted per CSV row, rotated to the track tangent.
6. Slope arrows rotate from G_TAIL / G_HEAD text: starting with **R** = 21 degrees, **F** = 339 degrees, containing **LEVEL** = 0.
7. Rows with a G_POSITION outside the polyline length are skipped and listed. The end report shows inserted and skipped counts.

Example CSV (from vidGradient_template.csv)

G_POSITION	G_CHAINAGE	G_RAILLEVEL	G_LINE	G_TAIL	G_HEAD	G_NOTE
0	0/000.000	1730.339			F 357	FOR EXIST - ALL LINES
240	0/240.000	1729.667		F 357	F 417	FOR EXIST - ALL LINES
320	0/320.000	1729.475		F 417	F 339	FOR EXIST - ALL LINES
400	0/400.000	1729.239		F 339	F 800	FOR EXIST - ALL LINES
540	0/540.000	1729.064		F 800	F 323	FOR EXIST - ALL LINES
640	0/640.000	1728.754		F 323	F 303	FOR EXIST - ALL LINES
720	0/720.000	1728.49		F 303	F 500	FOR EXIST - ALL LINES
800	0/800.000	1728.33		F 500	F 230	FOR EXIST - ALL LINES

Notes: Make the levels CSV with Rail Track Export (section 4), then fill the slope columns. Use Gradient Align and Clean Up afterwards if needed.

10. Gradient Align

`VIDGRADIENTALIGN`

Copies selected blocks into a neat staggered grid, for example to lay gradient blocks out as a sheet. Originals stay; copies get rotation 0.

Setup

- No file setup. Works in AutoCAD and Civil 3D. Select INSERT blocks (the gradient blocks).

How to use

1. Click **Gradient > Align** (or type VIDGRADIENTALIGN).
2. Fill the dialog (all fields accept 0; last values are remembered). Defaults shown in brackets.

Field	Meaning
Horizontal space (45)	X distance between blocks.
Vertical space (45)	Y distance between blocks.
Row count (3)	Blocks per column. 0 = no limit.
Column count (500)	Blocks per row. 0 = no limit.
Row offset (5)	X shift for offset rows.
Column offset (0)	Y drop applied per next column.
Offset step (2)	Offset cycle. 2 gives 0,1,0,1 ... and 3 gives 0,1,2,0,1,2 ... 0 = no offset.
Direction (Vertical)	Vertical fills down each column then the next column. Horizontal fills across each row then the next row.

1. After OK, **pick a point** (top-left of the layout, where block 1 goes).
2. **Select the blocks** to copy. They are ordered left to right (ties top to bottom).
3. Copies are placed. When Row count x Column count is filled, the next set starts directly below. The message reports blocks copied.

Notes: One Undo reverses the whole run.

11. Gradient Clean Up

`VIDGRADIENTCLEANUP`

Bursts gradient blocks so the attributes become plain MTEXT, ready for plotting or sending as basic geometry.

Setup

- No setup. Works in AutoCAD and Civil 3D. Does not need Express Tools.

How to use

1. Click **Gradient > Clean up** (or type VIDGRADIENTCLEANUP).
2. Select the gradient block(s).
3. Each block is exploded (burst). Every visible attribute becomes text, and all text from the burst is converted to MTEXT, keeping position, height, rotation, style, layer, colour and justification.
4. The command runs silently with no report. Undo reverses it.

Notes: This cannot be re-edited as a block afterwards, so keep a copy of the drawing. To limit it to blocks named G_NEW_FINAL, set VGC-OnlyGradient to T in the LSP.

12. Delete CAD Layers

`VIDCADMERGELAYERS`

Merges a fixed list of unwanted CAD / Civil 3D layers into layer 0, which clears them from the drawing. Useful after importing a DWG with many template layers.

Setup

- No setup. The layer list (A-BLDG, C-ANNO..., C-HYDR..., C-PROP..., V-... and so on) is inside the LSP file; edit `VIDCADMERGELAYERS:LayerList` to add or remove names.

How to use

1. Click **Delete Cad Layers > Delete** (or type VIDCADMERGELAYERS).
2. Layer 0 becomes current. Layers in the list that exist in the drawing are unlocked and thawed.
3. One -LAYMRG command merges all of them into 0. Layers not in the drawing are skipped.
4. The report shows layers merged and layers not found. Your previous current layer is restored.

Notes: Objects on those layers move to layer 0, so save a copy first.

13. Dim Settings

`VIDDIMSETTINGS`

Applies standard dimension settings with one click: text style font, arrow size, text height, text alignment and precision.

Setup

- The drawing needs a dimension style named **Standard** and a text style named **Standard**.

How to use

1. Click **Dim Settings > Apply** (or type VIDDIMSETTINGS).
2. The text style Standard gets font Arial. The dimension style Standard is updated and saved, then your previous dimension style is restored.

Setting	Value
Arrow size	1.5
Text height	1.5
Text alignment	Aligned with dimension line (inside and outside)
Primary units precision	0.00
Font	Arial

Notes: To change values, edit the VIDDim:Settings block at the top of vidDimSettings.lsp. If the text style has a fixed height, a note says dimension text uses that height instead.

14. Convert Dim Unit (meters to feet)

`VIDDIMMTOFT`

Changes the text of selected aligned or linear dimensions from meters to feet and inches, e.g. 3.92 becomes 12' 10".

Setup

- Drawing units must be **meters**.
- Works on aligned, rotated and linear dimensions only.

How to use

1. Click **Convert Dal unit > Convert** (or type VIDDIMMTOFT).
2. Choose an option (Enter = Override): **Override** shows feet only. **Prefix** shows feet on top and meters below. **Suffix** shows meters on top and feet below. **Reset** goes back to the measured value.
3. Select the dimensions. The command reports how many were updated.

Notes: Whole numbers only. Inches are always two digits and rounded, and 12 inches carries into feet.

15. CSV formats and troubleshooting

CSV columns each tool needs

Tool	Required columns	Optional
Survey Points	Point Name, Code, Northing, Easting, Elevation	Local Time
Rail Track Generate	Point Name, Code, Northing, Easting, Elevation	Local Time
OHE	Code, Northing, Easting	-
Building	Code, Northing, Easting, Elevation	Local Time
Rail Track Export (output)	Chainage, Elevation (written by the tool)	-
Gradient	G_POSITION, G_CHAINAGE, G_RAILLEVEL, G_LINE, G_TAIL, G_HEAD, G_NOTE	-

Header names are not case sensitive. Northing and Easting must be numbers in the drawing's coordinate system.

Suggested workflow

1. Load the tools and set the Map Zone.
2. Generate: Survey points, Rail track, Ohe, Building from the DGPS CSV.
3. Chainage on the track polylines.
4. Rail Track Export to make the level CSV, then Gradient Generate, Align and Clean Up.
5. Finish with Dim Settings and Delete CAD Layers.

Troubleshooting

Problem	What to do
Command not found	Run vidLoader.lsp again (APPLOAD) and read the [OK]/[ERROR] list.
Tools folder not found	Put the tools folder next to vidLoader.lsp, or pick logicDwg.lsp when asked.
Required headers missing	Check the CSV header names against the table above.
vid_blocks_tools.dwg not found	Keep it in the tools folder. Add the folder to Support Path and Trusted Paths.
Block imported but attributes empty	Check the block has the expected attribute tags (see Gradient and Chainage).
Gradient rows skipped	G_POSITION is longer than the selected polyline. Select the right track.
OHE finds no tracks	Draw tracks on layer 1-track first, or choose Select at the prompt.
Wrong map location	Use the correct Map Zone (43 or 44) before generating.